Yashaswini Ramakrishna

Senior Embedded Software Engineer | C/C++ | Yocto | Linux | RDK | Python
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Education

Visvesvaraya Technological University — Karnataka, India
Bachelor of Engineering - Electronics and Communication; Score: 84% — *June 2014 - June 2018*
Courses: Microcontrollers, Signals and Systems, Digital Signal Processing (DSP), VLSI Fundamentals, Microprocessors, Satellite Communication

Skills Summary

- **Languages** : C++, C, Python, bash
- **Embedded Development** : Yocto
- **Tools** : Bitbake , CMAKE and AutoTools, GIT (github/gitlab/bitbucket), gerrit, gdb, valgrind, MySQL, readelf, Jenkins, Jira
- **Multimedia Frameworks** : Gstreamer
- **Platforms** : Linux, Windows, Arduino, Raspberry, RDK

Experience

Tata Elxsi — Philadelphia, US
Senior Embedded Software Engineer (Full-time) — *November 2018 - Present*
Project client: **Comcast**

- **Development of OTT apps for RDK (C/C++)**:
  Led the design and development of premium apps porting layer of Amazon Prime, HBO Max, YouTube, and Netflix, for the RDK community. Key contributions include:
  - Developed and maintained the Netflix integration/device porting layer, implementing platform-specific changes in accordance with common port architecture and RDK standards.
  - Strong understanding of the Netflix application integration stack, including player/GStreamer components, DRM integration, authentication and authorization flows, and application-platform interfaces.
  - Owned end-to-end Netflix certification, including certification test execution, test-case analysis, issue investigation, debugging, bug fixing, fix validation, and coordination with dependent platform/SoC teams.
  - Worked extensively with the Netflix Testing Suite (NTS) for application validation, certification, and troubleshooting.
  - Performed EyePatch and test-case analysis to understand application behavior, identify failure points, and debug certification issues across the application and platform layers.
  - Worked on integrating and validating new Netflix features and capabilities, including Cloud Gaming and Multiplayer.
  - Developed and maintained Netflix launcher integration on RDK-based platforms.
  - Working knowledge of other major native OTT applications, including Amazon Prime Video and YouTube, and their integration requirements within the RDK ecosystem.

- **Reference Home UI for RDK Video Devices (Lightning/JavaScript)**:
  - Solid hands-on experience working with Reference Home UI for RDK Video Devices
  - Redesigned the RDK UI/UX, a JavaScript- and Lightning-based application, to support adoption of the DAC (Downloadable Application Container) architecture.
  - Oversaw UI development activities across implementation, integration, and validation, ensuring alignment with DAC architecture and RDK platform requirements.
  - Strong understanding of application lifecycle management through the Reference Home UI, including application discovery, launch, and management.
  - Hands-on experience with DIAL (Discovery and Launch) integration and its interaction with the RDK Reference Home UI.
  - Strong understanding of Device Automation Bus (DAB) and its integration with the Reference Home UI, including UI changes required to support device automation and validation.

- **Provisioning & Cryptography for Native Apps (C/C++)**:
  - Strong understanding of cryptographic APIs and security mechanisms used by native OTT applications for authentication and provisioning, including foundational cryptographic operations such as encryption, HMAC, and Diffie-Hellman key exchange.
  - Contributed to the development of the iCrypto Thunder plugin, supporting cryptographic requirements for OTT application authentication and provisioning.
  - Developed a cryptography abstraction layer to enable support for multiple crypto providers and implementations, including OpenSSL, SecAPI, and direct SoC-level crypto calls.
  - Supported the integration of SoC Trusted Application (TA) libraries for cryptographic operations and security requirements, primarily for Netflix integration.

- **DRM, OCDM & Thunder Plugins (C/C++)**:
  - Strong understanding of DRM architecture and content protection, including OCDM (Open Content Decryption Module), Widevine, PlayReady, and Secure Video Path (SVP), and their integration within the RDK ecosystem.
  - Hands-on experience with DRM integration and troubleshooting across the application, OCDM, and underlying platform/SoC layers.

- **RDK Thunder plugins (C/C++)**:
  - Hands-on experience developing Thunder plugins for RDK, including development of Continue Watching plugin.

Leadership & Community Engagement

Tata Elxsi Young Leaders Programme (YLP) Philadelphia,US
2024-2025

* **Leadership Development**: Selected for and successfully completed Tata Elxsi's Young Leaders Programme (YLP), a leadership development initiative focused on preparing high-potential employees for future leadership roles.
* **Skills Development**: Developed skills in communication, critical thinking, negotiation, decision-making, and delegation through structured learning, mentorship, and simulation-based assessments using KNOLSKAPE, involving real-world business scenarios and leadership decision-making.

Rotaract Club Banglore, India
Active Member | Club Secretary

* **Community Engagement**: Active member of the Rotaract Club, affiliated with Rotary International District 3190, during college.
* **Club Leadership**: Served as Club Secretary for a club with approximately 500 members, supporting club operations, coordination, communication, and member engagement.
* **Event Management**: Planned and organized college-level events, coordinating with teams and managing activities across the event lifecycle.